

Incorporating environmental change over succession in an integral projection model of population dynamics of a forest herb

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Despite seemingly obvious effects of environmental drivers, mechanisms behind long-term changes in plant population sizes over time are often poorly known. We investigated how soil potassium concentration and seed predation are likely to change over time as a result of succession from deciduous forest to spruce forest, and how this affects population trajectories of *Actaea spicata*. Observations and addition experiments showed that high soil potassium concentration increased individual growth rates. Among-site comparisons showed that soil potassium concentration was lower where proportion spruce was higher. Incorporation of a gradual increase in spruce over time in an integral projection model where individual growth depended on potassium suggested a net decrease in *A. spicata* population sizes over forest succession. This result suggests that small changes in factors with small effects on individual performance can influence patterns of species occupancy along successional gradients. We incorporated also density independent and density dependent effects of pre-dispersal seed predation over succession into the same model. Seed predation influenced the tree composition at which *A. spicata* population growth was positive. However, significant effects of *A. spicata* population size on seed predation intensity did not translate into important feedback effects on population growth trajectories over succession. Our results illustrate how demographic models can be used to gain understanding of the mechanisms behind effects of environmental change on species abundances and distributions by the simultaneous inclusion of changing abiotic and biotic factors.

Natural environments change over time due to succession, eutrophication, altered land-use, climate change or other causes. This potentially has large consequences for population dynamics and viability of many organisms, and many species that are rare or threatened occur in environments that undergo change (Caughley 1994, Norris 2004). Indeed, among-population comparisons suggest that variation in environmental factors have substantial effects on plant population dynamics (Alvarez-Buylla 1994, Pascarella and Horvitz 1998, Gotelli and Ellison 2002), which should entail effects on abundances and distributions (cf. Clark 2009; Fig. 1). These observations notwithstanding, directional temporal environmental change is usually not incorporated explicitly in demographic models or in population viability analyses (Caswell 2001, Morris and Doak 2002, but see Gotelli and Ellison 2006, Dahlgren and Ehrlén 2009). One reason for this is that we largely lack understanding of how environmental factors influence demographic parameters and how they change over time. While temporal environmental variation is considered essential for realistic models of population dynamics (Tuljapourkar 1989, Morris et al. 2008), it is usually incorporated only implicitly, in terms of stochastic variation in vital rates (Morris and Doak 2002, but see e.g. Smith et al. 2005). Explicit inclusion of temporal variation in environmental factors, and how this affects vital rates, is important to increase accuracy

of population models for fluctuating environments (Fieberg and Ellner 2001), and indispensable to realistically model population dynamics in directionally changing environments (cf. Gotelli and Ellison 2006, Dahlgren and Ehrlén 2009). From a conservation perspective, models linking population dynamics to manageable environmental factors will be essential to identify optimal management actions for threatened species in changing environments.

Temporal environmental change in terms of succession is considered to have important effects on species abundances and community composition in many systems (Connell and Slatyer 1977, Pickett et al. 2010). Community change over succession may be regarded as both a cause and a consequence of changes in the abiotic environment (Gleason 1926). However, to individual species this dual relationship may be highly asymmetric. For example, forest floor species experience changes in light availability, soil water content and soil nutrient concentrations over succession that are largely results of changes in the density and composition of overstory trees (Schwendenmann 2000, Hart and Chen 2006, but see Royo and Carson 2006). In addition to the abiotic environment, the intensity of biotic interactions of forest floor species such as herbivory and seed predation may change during succession (Fagan and Bishop 2000, Crist et al. 2005). Changes in biotic interactions during succession may partly be a result of changes in population densities of the interacting plants

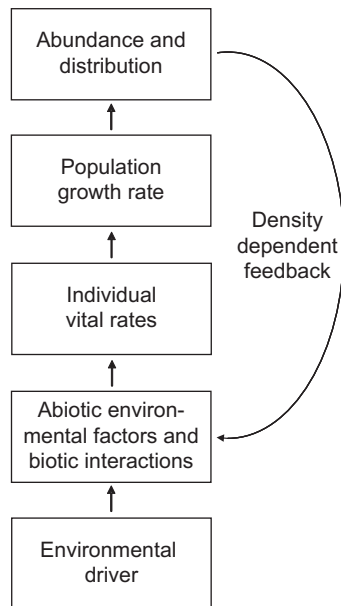


Figure 1. A conceptual model of how environmental drivers, such as succession, can affect species abundances and distributions through environmental effects on demography. After Clark (2009).

(Halpern and Underwood 2002). Biotic interactions may then feed back on the growth of plant populations, counteracting or reinforcing effects of changes in the abiotic environment. A mechanistic understanding of population dynamics of individual species over succession thus requires that we can link changes in abiotic environmental factors to population growth rate also via effects on biotic interactions that influence vital rates.

We combined information about how abiotic and biotic environmental factors affect vital rates of the herb *Actaea spicata* with how these factors are likely to change over succession from deciduous to spruce forest, in order to examine whether resulting effects on population growth can explain the lack of *A. spicata* in late successional forest. We used information from a previous correlative study, where soil potassium concentration and pre-dispersal seed predation were identified out of a set of environmental variables to affect vital rates of *A. spicata* (Dahlgren and Ehrlén 2009). To corroborate the effects of potassium, acting via individual growth, we performed additional observations and an experiment. We used a chronosequence of forest patches to examine how soil potassium changes during succession. We then included temporal change in potassium in an integral projection model to predict effects of changing potassium on population size. We also included environmental stochasticity in this model. In addition, we specified models examining how seed predation influences population trajectories over succession via density dependent and density independent effects, based on previous observations on seed predation in *A. spicata* populations of different sizes (von Zeipel et al. 2006). We asked: (1) does decreasing soil potassium concentration over forest succession partly explain the fact that *A. spicata* populations usually are small or missing in late successional spruce forest? (2) Does seed predation counteract or reinforce effects of changes in soil potassium over succession on population growth rate? (3) Does density dependent seed

predation have feedback effects on growth trajectories of *A. spicata* over succession?

Material and methods

Study species and area

Baneberry *Actaea spicata* (Ranunculaceae) is a long-lived perennial toxic herb with black fleshy fruits. In northern Europe it occurs in shady, well-drained and nutrient rich patches under deciduous and mixed stands with deciduous trees and spruce (Pellmyr 1984). It may sometimes be found in nutrient rich patches of spruce-dominated forests but is typically missing in spruce forests and in patches with few trees, corresponding to later and earlier successional stages, respectively (Mossberg and Stenberg 2003, Dahlgren unpubl.). Previous studies have documented higher population growth rates for *A. spicata* in deciduous forest than in mixed coniferous forest (Fröborg and Eriksson 2003). In southeastern Sweden, forest succession constitutes a threat to *A. spicata* due to a regional loss of suitable habitat. Individuals can produce several shoots, each bearing up to four infructescences (Eriksson 1995). Infructescences bear 1–30 fruits, typically containing 8–16 seeds each, and fruit set is close to 100% (von Zeipel et al. 2006). Seeds germinate below ground one year after release and aerial parts are produced one year thereafter. High soil potassium concentration has previously been shown to increase growth of individuals in greenhouse experiments (Salomonson et al. 1992). A positive correlation between potassium and individual growth rate has also been observed in the field, in demographic models leading to higher population growth rates in potassium rich patches (Dahlgren and Ehrlén 2009). Fruits are frequently infested by a specialist pre-dispersal seed predator, *Eupithecia immundata* (Geometridae). Oviposition occurs early in the season, almost exclusively affecting the early developing primary infructescences (Eriksson 1995). Typically one larva is developed per fruit, leaving on average less than one intact seed per attacked fruit (Dahlgren and Ehrlén 2009). Seed predation intensity is curvilinearly related to host plant population size, with the highest predation in medium to large sized populations because of a lack of *E. immundata* in smaller populations and the presence of several parasitoids attacking *E. immundata* in the largest populations (von Zeipel et al. 2006).

In a previous study of *Actaea spicata*, nine factors were tested for correlations with vital rates in addition to soil potassium concentration and seed predation (Dahlgren and Ehrlén 2009). These factors were intra- and interspecific density and seven measures of abiotic factors (ground slope, soil depth, May canopy cover, July canopy cover, soil phosphorus, soil water content, and soil temperature). No effects of density were found, and out of the measures of abiotic factors, only an effect of May canopy cover on fruit number was identified. The effect of May canopy cover on population growth rate was weak in comparison to the effect of potassium. Thus, we only examine effects of soil potassium and seed predation in the present study, and use population models that are density-independent except for effects of population size on seed predation.

Field studies were conducted in the forests of Tullgarn natural reserve, 45 km SSW of Stockholm, Sweden (58°6'N, 17°4'E). Tree composition ranges from deciduous stands dominated by birch, oak, ash, maple, lime, aspen and hazel to coniferous stands dominated by Norway spruce *Picea abies*. Forest patches are interspersed with agricultural land. Historically the majority of forest patches were used as wooded meadows, containing many deciduous tree species valuable for agriculture because of fertilizing effects and other uses, such as the use of leaves as winter fodder for cattle (Ryberg 1971). After abandonment, succession has gradually transformed former meadows and deciduous woodlands into forests with increasing proportions of spruce over time. Today most parts of the forests are dominated by spruce. *Actaea spicata* is abundant in the study area. The population size distribution is skewed with a mean of 360 and a range of 4–20 900 established non-seedling individuals (von Zeipel 2007, Dahlgren unpubl.).

Data collection and analysis

In order to corroborate correlative results of positive soil potassium effects on individual growth, we experimentally added potassium to 12 of the 25 plots included in the demographic study described below. Treatments were applied randomly at the end of the growing season in 2007, after data collection for the descriptive study was finished. The remaining 13 plots were kept as controls. Thirty grams fertilizer per square meter was applied evenly to each treatment plot. The fertilizer was specified as containing 30.7% potassium as the sole cation nutrient. The potassium was encapsulated within granules with slowly biodegradable soy-based coats, leading to a gradual release. Individuals in treatment and control plots were censused in 2008 and in 2009. For statistical tests of potassium effects, the 'lmer' function in the 'lme4' package for R 2.9.1 (R Development Core Team 2009) was used to fit a multilevel mixed-effects model. In this model, size was a function of size the previous year that varied between individuals, and the potassium treatment, that varied between plots, in the presence of random plot and site effects. Conservative p-values were calculated using MCMC-methods and the 'pvals.fnc' function in the 'languageR' package.

Data to estimate changes in potassium concentration over succession from deciduous to spruce forest was collected from a chronosequence of forest locations in the study area. Soil samples were collected in 2008, just before shoot emergence of *A. spicata*, at 100 locations with tree composition ranging from pure deciduous to pure spruce. The sample locations included the four *A. spicata* patches of the demographic study. Ninety-six additional locations were evenly distributed in a 3 × 4.5 km area in the surrounding forests, using land-use maps. Exact locations were chosen in the field so that replicates constituted either distinct stands or were situated so that locations were more than 50 m from other locations and of different tree composition than locations within 100 m (Dahlgren unpubl.). At each location, nine soil samples evenly spaced over a 10 × 10 m flat area were taken. Samples were collected outside the plots in the demography patches. Soil samples were pooled for analysis and potassium concentration was determined using the AL

extraction method (Salomon 1998). In 2009, all locations were revisited and the fraction of tree density constituted by spruce (hereafter proportion of spruce) was estimated using an 'angle gauge', a forestry instrument used to estimate basal area of trees. The relationship between soil potassium concentration and proportion of spruce was modeled with linear regression.

Seed predation has previously been shown to covary with host plant population size in the study system (von Zeipel et al. 2006, von Zeipel 2007), suggesting a density dependent relationship. In order to quantify this relationship, we recalculated number of flowering individuals reported in von Zeipel (2007) to number of established (non-seedling) individuals by multiplying the original values with the ratio of established to flowering individuals in our study. We also recalculated the proportion of preyed fruits in primary infructescences to the proportion total fruits preyed, by multiplying original seed predation estimates by the proportion of fruits occurring in primary infructescences. We then fitted a second-order polynomial regression line to the relationship between number of established individuals and proportion of fruits exposed to seed predation, using pooled data over all four years analyzed in von Zeipel (2007) (Appendix 1). We only used the increasing relationship between population size and predation from small to large populations in further analyses, because the large number of observations available for this range resulted in more precise estimates.

The demographic model

Demographic data was collected in 25 plots of 5 × 5 m at four sites each year from 2004 to 2007 for a total of 1543 unique individuals. Sites were heterogeneous and chosen to include plots representative of *A. spicata* habitat patches in the area with regard to tree type and cover. Distances between sites varied from 0.8 to 2 km. Individual size and fruit number was measured each year for all established individuals, and all seedlings were counted. Fruit predation was recorded each year as number of fruits per individual with exit holes of larvae. Soil potassium concentration was measured in 2005, in a pooled soil sample for each plot quadrant. Individuals were assigned a unique potassium value as a weighted average based on the distance to the sampling points in each plot quadrant center. The positive effect of a high soil potassium concentration on individual growth (but lacking effects on survival, probability of flowering and fruit number) was determined with mixed-effects regression analyses, taking the spatial and temporal structure and additional environmental variables into account. Linear regressions of size over size the previous year and potassium were performed separately for 2005, 2006 and 2007 in order to obtain yearly estimates of effects of potassium on individual growth for parameterization of a stochastic integral projection model. We did not measure within- and among-year fluctuations in soil potassium since our main aim was to determine relative concentrations between individuals.

To incorporate effects of changing potassium concentration and seed predation over succession into size structured models of population dynamics we used integral projection models (IPMs). IPMs resemble matrix projection models, but instead of a projection matrix, a continuous projection kernel

is used and the population size distribution is described by a density function instead of a vector (Easterling et al. 2000). Growth, survival and fertility components are included in the IPMs based on regression models with one or several predictor variables, using all sampled individuals for parameterizing each vital rate for a given state (Ellner and Rees 2006). Our model consisted of three coupled equations, allowing a delay of seedling establishment and treatment of seeds and seedlings as discrete classes (cf. Rees et al. 2006, Dahlgren and Ehrlén 2009):

$$n(y, t + 1) = p_s p_d(y) S_2(t) \int_{\Omega} s(x) g(y, x, z) n(x, t) dx \quad (1)$$

$$S_2(t + 1) = p_e S_1(t) \quad (2)$$

$$S_1(t + 1) = \int_{\Omega} p_f(x) f_n(x) f_s n(x, t) dx \quad (3)$$

where n is a distribution function describing the adult population, S_2 is the number of seedlings, and S_1 is the number of seeds. x is size year t and y is size year $t + 1$. Ω delimits the range of possible sizes. p_s , p_d , and p_e are probability of seedling survival, the size distribution of established seedlings, and probability of seedling establishment, respectively. Potassium (z) was included as a second predictor variable in the linear regression of y on x describing individual growth (g). The residual variation in the regression model was used to obtain a probability distribution of y . s and p_f are functions of survival and flowering, respectively, estimated by logistic regression. f_n is fruit number, estimated by Poisson regression. The number of seeds per fruit (f_s) depended on predation rate (p_p) and mean number of seeds in intact (m_i) and preyed fruits (m_p) according to:

$$f_s = m_p p_p + m_i(1 - p_p) \quad (4)$$

All parameters are given in Table 1 (for model details see Dahlgren and Ehrlén 2009).

Simulations

We parameterized IPM kernels for three transition intervals; 2004–2005, 2005–2006 and 2006–2007 (Table 1). We then formulated a stochastic IPM where one transition kernel was drawn randomly at each time-step, and ran it in R 2.9.0.

The model was used to calculate stochastic population growth rates for *Actaea spicata* at potassium concentrations corresponding to either pure deciduous forest or pure spruce forest at different levels of seed predation. Calculations were made numerically, as the geometric mean of annual growth rates in iterations of the model for 10 000 (disregarding 1000 initial) time steps (years). We also modeled population dynamics of *Actaea spicata* populations experiencing changing potassium concentrations over time due to succession from deciduous to spruce forest. In these simulations we compared models with observed seed predation, with no seed predation, and with seed predation intensity depending on *Actaea spicata* population size. For the succession scenario, we let potassium concentration in the IPM change gradually from deciduous forest values to spruce forest values according to the regression analysis of the chronosequence data. We multiplied the chronosequence estimates by the mean ratio of concentrations in 2005 to 2008 in the demography patches, as differences in potassium between the two years in the demography patches used to parameterize the integral projection model were likely caused mainly by within-year fluctuations. We assumed the starting point to be an old deciduous forest, in which aging trees were disappearing and being replaced by spruce. This replacement was assumed to take 120 years and correspond to a sigmoid increase in spruce basal area over time (Chertov et al. 1999). Assuming a linear increase in spruce basal area over time resulted in qualitatively similar outcomes of the simulations (not shown). Density dependent changes in seed predation were implemented in the IPM according to the relative relationships between seed predation intensity and population size in the regression analysis. The presented simulations started with 50 established individuals, corresponding to the observed median population size in the study area, but the results were similar also with other initial population sizes (not shown). The succession scenarios were iterated 10 000 times. The initial size distribution was set as the stable distribution of a deterministic model with a mean transition kernel from the three transition intervals and potassium concentration corresponding to deciduous forest. In order to account for the uncertainty in the relationship between soil potassium and spruce proportion we also ran simulations where the slope of potassium was first increased and then decreased by 1.96 times its standard error (corresponding to the 95%

Table 1. Components and parameter values of the integral projection model for *Actaea spicata* individuals in SE Sweden 2004–2007. y = size year $t + 1$, x = size year t , z = soil potassium concentration, p_p = proportion of fruits exposed to predation.

Model component	Value 2004–2005	Value 2005–2006	Value 2006–2007
Survival	$\text{logit}(s) = -1.73 + 0.65x$	$\text{logit}(s) = -2.98 + 0.54x$	$\text{logit}(s) = -1.69 + 0.51x$
Growth	$y = 1.62 + 0.82x + 0.0080z$ $\sigma^2 = 1.25$	$y = 2.42 + 0.68x + 0.011z$ $\sigma^2 = 0.98$	$y = 1.70 + 0.72x + 0.015z$ $\sigma^2 = 1.43$
Flowering	$\text{logit}(f_p) = -8.56 + 1.02x$	$\text{logit}(f_p) = -12.62 + 1.48x$	$\text{logit}(f_p) = -12.97 + 1.53x$
Fruit number	$\ln(f_n) = 0.69 + 0.25x$	$\ln(f_n) = -0.77 + 0.39x$	$\ln(f_n) = -2.62 + 0.59x$
Seed number ¹	$f_s = 0.39p_p + 9.26(1 - p_p)$	$f_s = 0.39p_p + 9.26(1 - p_p)$	$f_s = 0.39p_p + 9.26(1 - p_p)$
Germination ²	$p_e = 0.0041$	$p_e = 0.014$	$p_e = 0.018$
Seedling survival	$p_s = 0.21$	$p_s = 0.27$	$p_s = 0.25$
Size of surviving seedlings ³	$p_d = N(3.31, 1.80)$	$p_d = N(2.84, 2.43)$	$p_d = N(3.94, 0.81)$

¹the model is the same for all transition intervals, values refer to average seed number in attacked and intact fruits in 2005.

²the probability of germination includes the probability of seed survival. Values were calculated based on observations of fruit numbers in years 2004, 2005 and 2006, and seedling numbers in years 2006, 2007 and 2008.

³ $N(\mu, \sigma^2)$ denotes a normal distribution with mean μ and variance σ^2 .

confidence limit). We did not incorporate the variability in the potassium–spruce relationship as stochastic variation in the model because this variability likely reflects mostly spatial, not temporal, variation.

Results

Experimental potassium addition in 2007 increased growth of *Actaea spicata* individuals from 2008 to 2009 (main effect: $b = 1.30$; $p = 0.020$ based on the posterior distribution of 50 000 MCMC iterations). The effect was stronger for smaller individuals (interaction effect between size previous year and potassium: $b = -0.14$; $p = 0.034$) and was predicted to be close to zero for the largest observed individuals.

Among stands, potassium concentration decreased with proportion spruce (Fig. 2). It was statistically significant, but there was a lot of scatter (spatial variation) in the relationship. Thus, succession from deciduous forest to spruce forest should result in a decrease in potassium over time although there is a considerable variation among sites with similar proportion of spruce.

The effects of potassium on individual growth were similar over the three study years (Table 1). The estimated slope coefficients of potassium concentration in the multiple regression models describing individual growth were 0.008 ($p = 0.012$) for 2004–2005, 0.011 ($p < 0.001$) for 2005–2006 and 0.015 ($p < 0.001$) for 2006–2007. For individuals of the mean observed size, the regression models predicted a mean growth of 7.3% in deciduous forest and 6.3% in spruce forest for the 2004–2005 transition, 5.3% in deciduous and 4.0% in spruce forest for 2005–2006, and 2.0% in deciduous and 0.2% in spruce forest for 2006–2007. There were no significant interactions between size and soil potassium concentration. There was little scatter in the correlation of size with size the following year in all transitions. Size also had strong effects on survival and reproduction. Survival of established

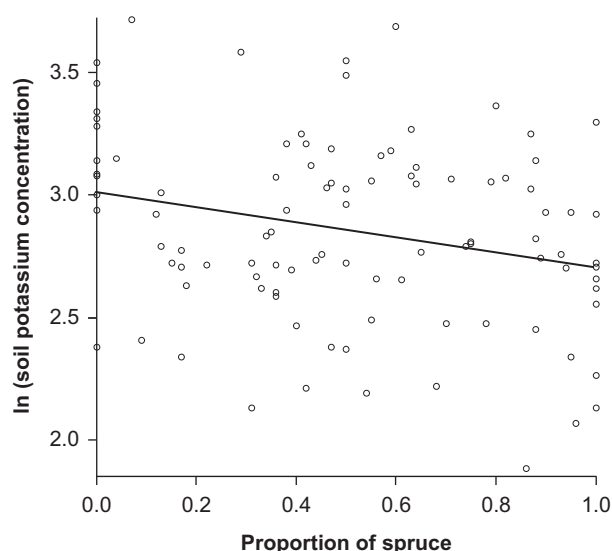


Figure 2. Regression of soil potassium concentration (natural logarithm of mg potassium/100 g soil) on proportion of spruce ($b = -0.30$, $SE = 0.12$, $p = 0.01$, $R^2 = 0.07$). An extreme outlier that made the relationship stronger was removed.

individuals was high overall, but markedly higher between 2004 and 2005 (0.98 for the most common size) than the other years (mean = 0.91). Mean flowering and fruit number were similar between years but relationships with size varied in magnitude. Proportion of fruits exposed to predation was 10% in 2004, 29% in 2005 and 27% in 2006.

At potassium concentrations observed in deciduous stands, and with density-independent seed predation randomly varying between the three observed yearly rates, *A. spicata* populations were predicted to remain at constant size or increase slightly (stochastic $\lambda = 1.001$). At concentrations observed in spruce stands, populations decreased (stochastic $\lambda = 0.982$). In the simulation of succession from deciduous to spruce forest over 120 years average populations remained almost constant until 12% of trees were spruce, corresponding approximately to 33 years according to the assumed sigmoid-shaped increase in spruce basal area (Fig. 3). At later stages of succession and higher proportions of spruce, populations started to decline as a result of decreasing soil potassium concentration.

Seed predation changed the predicted population trajectory during forest succession. In an integral projection model with no seed predation, initial increases were more pronounced and populations started to decline later during succession (Fig. 4). The stochastic population growth rate in a model with no seed predation and with deciduous forest potassium concentrations was 1.014. With no seed predation and with spruce forest potassium concentrations growth rate was 0.993. Despite a small change in λ , there were quite large differences in population trajectories between models with and without seed predation as a result of that the model without seed predation yielded growth rates above one for many forest types where growth rates were below

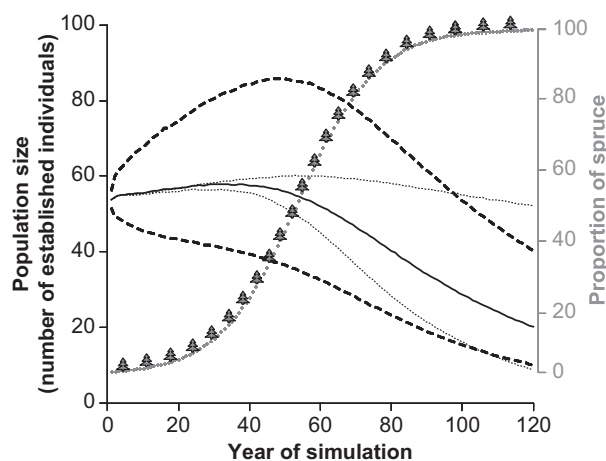


Figure 3. Simulated population growth of *Actaea spicata* over 120 years of changing soil potassium concentration during succession from deciduous to spruce forest. The solid line is the geometric mean population size of 10 000 iterations of the integral projection model (non-seedling established individuals). The dashed lines represent the uncertainty resulting from observed between-year variation, in terms of the geometric mean population size from the model, multiplied or divided by one geometric standard deviation. The dotted black lines show the 95% CI of the mean predicted population trajectory resulting from the uncertainty in the relationship between soil potassium concentration and proportion of spruce. Proportion of spruce is in gray.

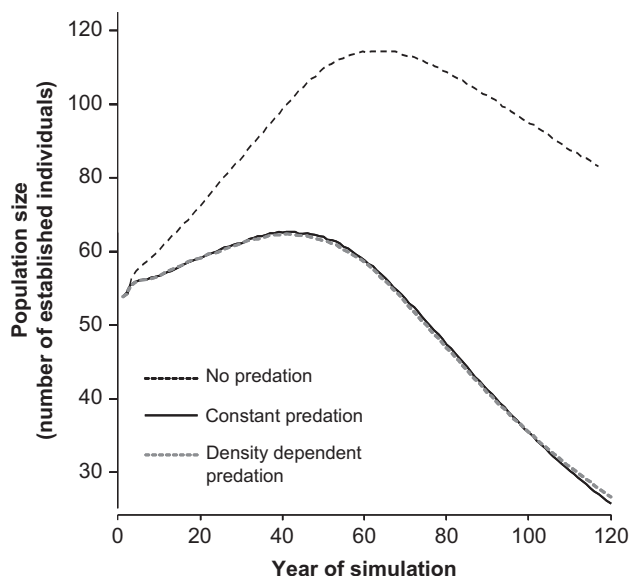


Figure 4. Simulated population growth of *Actaea spicata* over succession from deciduous to spruce forest in models with no seed predation, seed predation rate constant at the level predicted for the initial population size, and seed predation rate changing with host population size.

one in models with seed predation. Population size was non-linearly related to seed predation (proportion of fruits preyed = $-0.10 + 0.40 \times \log_{10}(\text{population size}) - 0.07 \times \log_{10}(\text{population size})^2$, $F_{2,442} = 48$, $p < 0.0001$). Still, in contrast to the marked difference between models with and without seed predation, models with plant population size dependent vs. independent predation intensities differed very little (Fig. 4). This was mainly the result of that differences in population size caused by potassium were small. Larger differences in population size were predicted to lead to substantial differences in seed predation (for example, in a population with ten established individuals seed predation was estimated to be 51% of the intensity in a population with 100 established individuals), and thus stronger effects on population growth rate.

Discussion

We examined how abiotic and biotic interactions that affect vital rates of *Actaea spicata* are likely to change during succession from deciduous forest to spruce forest, and if the resulting effects on population dynamics can explain why *A. spicata* is missing in late successional forest. Observational and experimental data confirmed that high soil potassium concentrations increase individual growth rates in *A. spicata*. In addition, we found that soil potassium concentrations are likely to decrease over forest succession since concentrations decreased with increases in proportion of spruce. Incorporation of the effect of potassium on individual growth and its change over succession in an integral projection model resulted in population trajectories that first increased and then decreased. However, the effect of potassium was not large enough to in itself explain why *A. spicata* populations are small or missing in late successional stages. Seed predation reduced population growth rates of *A. spicata* at all stages of succession, but feed-backs on seed predation from host plant population size had small effects.

The results suggest that temporal changes in soil potassium concentration partly explain the restriction of *A. spicata* to earlier-successional forest due to effects on individual and population growth rates. Although the predicted change in average concentrations of potassium during forest succession was small compared to variation among sites with similar forest composition, the relationship between proportion spruce and potassium concentration was significant. This is in agreement with that leaf litter nutrient content is known to differ between deciduous trees and spruce (Johansson 1986). Moreover, soil nutrient levels are known to change during forest succession also in other systems (Leuschner and Rode 1999, Yemshanov and Perera 2002). In the demographic model, average sizes of *A. spicata* populations were predicted to decrease during succession as a result of the soil potassium decrease. The facts that potassium concentrations varied among sites within successional stages, and that the rate of the replacement of deciduous trees with spruce is also likely to vary, both suggest that patterns of successional change vary between sites (Rammig et al. 2006). The fact that only a minor proportion of variation in vital rates was explained by potassium concentrations suggest that several other factors may be important for plant performance in this environment. The light environment likely sets the ultimate limits of the niche of *A. spicata* as it is never found in completely unshaded or very densely shaded patches. However, the effect of differences in canopy cover of the magnitude observed in natural populations has been shown to have a smaller effect on the demography of *A. spicata* than potassium concentration (Dahlgren and Ehrlén 2009). Our results suggest that small changes in an abiotic factor (potassium) other than light have the potential to drive changes in field layer composition over forest succession (cf. Barbier et al. 2008). Gotelli and Ellison (2002) found that small increases in nitrogen deposition rate substantially increased extinction risk in the pitcher plant *Sarracenia purpurea*. Taken together these studies illustrate that small changes in factors with low impact on population growth rate can have substantial effects on long-term population dynamics.

We modeled density independent population dynamics since our previous analyses have provided no evidence of effects of either intra- or inter-specific density on vital rates in natural populations. Still, because only experimental manipulations can reliably evaluate effects of density, we can not fully exclude the possibility that such effects indeed occur (experiments have been initiated in the study populations). To the extent that there is negative intraspecific density dependence in this system, the effect of a decrease in soil potassium on population growth rates over succession should be weaker than in the situation without density dependence, and if potassium ceases to decrease population sizes could stabilize at lower densities.

Our simulations suggest that biotic interactions like pre-dispersal seed predation influence changes in plant field layers over succession via effects on local population dynamics, also in long-lived species that are less sensitive to effects on reproduction. In the absence of seed predation λ would be about 0.01 higher than with seed predation for an average population, making a net decrease in population size over the simulated succession from deciduous to spruce forest less probable. For *A. spicata* in the study area, experimental

manipulations of seed densities do not result in compensating responses in seed and seedling mortality (Ehrlén and Eriksson 2000, Fröborg and Eriksson 2003, von Zeipel 2007). This suggests that reduced seed number indeed translates into proportional reductions in recruitment. Given that negative effects of intraspecific density are absent also during later life-stages, it should translate also into the effects on population growth rate predicted by the model. Herbivores have previously been shown to affect plant population growth during succession in other species (Brown 1984) and are known to affect realized niches, e.g. for serpentine ecotypes of *Collinsia sparsiflora* (Lau et al. 2008). The higher growth rate in the absence of seed predation in our simulations suggests that spruce dominated forests devoid of seed predators may constitute a habitat for *A. spicata* even though abiotic environmental conditions are suboptimal. In contrast to the lack of evidence for direct intraspecific density dependence, there is evidence of effects via seed predation, in terms of the covariation between *A. spicata* population size and pre-dispersal seed predation intensity. However, our simulations suggest that feedback loops via host plant population size and seed predation have very little effect on population trajectories over succession.

The explicit inclusion of environmental effects on vital rates make models of population dynamics and population viability analyses more mechanistic and useful in guiding management actions for threatened species. Including also temporal changes in the environment into demographic models is essential to study effects on populations of processes such as succession, eutrophication, land-use change, and climate change. Population sizes over time can then be described as functions of manageable factors, making models useful for designing optimal management of threatened and economically important species in changing environments. Although we did not find any evidence of density dependence in our system, such effects are likely to be manifested in some systems, and should therefore be considered in analyses. Generally, combined changes in a set of environmental factors resulting in long-term population growth rates below unity imply that the environment has ceased to constitute a possible habitat. Thus, models explicitly including changing environmental factors can be used also to determine how these factors restrict successional niches. Using similar approaches, future studies will hopefully be able to shed more light on the links connecting drivers of environmental factors and biotic interactions with species abundances and distributions via individual vital rates and population growth rates.

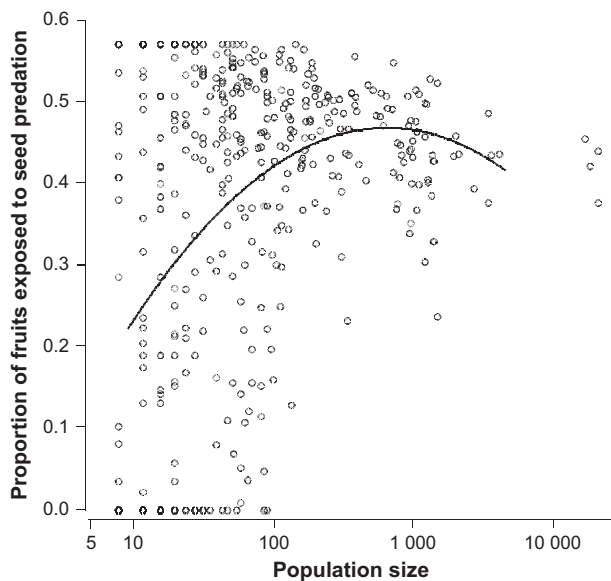
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Appendix 1. Proportion of fruits exposed to seed predation as a function of number of established *Actaea spicata* individuals. Data from 149 populations over four years are pooled and a second-order polynomial regression line is fitted. The maximum of 0.57 is the estimated proportion of fruits occurring in the attacked, primary, infructescences. Modified from von Zeipel (2007).